- How about the case with more than 2 periods?

For illustration, suppose we have *n* individuals and 3 periods.

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it} = \beta_0 + \beta_1 x_{it} + a_i + v_{it}, \qquad i = 1, \ldots, n; t = 1,2,3$$

- The first differenced equation is

(t=2) − (t=1): $y_{i2} - y_{i1} = \beta_1(x_{i2} - x_{i1}) + (v_{i2} - v_{i1})$

(t=3) − (t=2): $y_{i3} - y_{i2} = \beta_1(x_{i3} - x_{i2}) + (v_{i3} - v_{i2})$

$$\Delta \mathrm{y}_{it} = \alpha_0 + \beta_1 \Delta x_{it} + \Delta v_{it} \; i = 1, \ldots, n; t = 2,3$$

In case $\boldsymbol{E}(\Delta v_{it}) \neq 0$

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For illustration, suppose we have *n* individuals and 3 periods.

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$$\Delta \mathrm{y}_{it} = \alpha_0 + \alpha_1 time\_dummy + \beta_1 \Delta x_{it} + \Delta v_{it} \; i = 1, \ldots, n; t = 2,3$$

Use 1 dummy to differentia two groups (t=2 and 3)